

Part 2: SQL and PL/SQL (15 marks)

Bicycle Rental

The following tables: Bikes, Reservation, and Client are used by Bicycle Reservation System. The business rules are: (1) A bicycle may be rented for a half day or a full day (2) A bike can be rented only twice per day, and (3) **A bike is charged for a full day, if the bike is rented for more than 6 hours, and charged half day otherwise**

Bike <u>B_Id</u> B_color B_seater Cost_per_day	Reservation <u>Res_Id</u> Start_time (0-23) End_time (0-23) Date_of_rent date Client_Id B_Id Total_cost	Client <u>Client_Id</u> DOB C_Name C_Address C_City
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Use the schema given to create and populate tables.

Step 1: Add 2 bikes, 2 clients and 4 more reservations.

Step 2: Create and execute the following procedure and function.

A. Create a function, CalcCost() – that would calculate and return the total cost of the booking for the rental bicycle travel based on the res_id. The end_hour should be taken as a parameter. You need to get v_id from res_id and use it to get cost_per_day for the bike. Calculate number of hours of reservation. If number of hours are 6 or more, use full day charges. Now use the update command to finally put the total cost in the reservation table and change end_hour to time entered. You need to make sure start_hour is less than end_hours and nothing exceeds 23. [4 marks]

```
CREATE OR REPLACE FUNCTION CalcCost
(end_hour NUMBER, res_id NUMBER)
RETURN NUMBER IS
```

```
V_ID CHAR;
cost_per_day NUMBER;
hoursBetween NUMBER;
starttime NUMBER;
endtime NUMBER;
total NUMBER;
```

```
BEGIN
```

```
SELECT START_TIME INTO starttime FROM reservation01 WHERE R_ID = res_id;
```

```
SELECT B_ID INTO V_ID FROM reservation01 WHERE R_ID = res_id;
SELECT Cost_per_day INTO cost_per_day FROM reservation01 WHERE R_ID = res_id;
```

```
hoursBetween := end_hour - starttime;
```

```
IF hoursBetween >= 6 THEN total := cost_per_day;
ELSE total := (cost_per_day / 24) * hoursBetween;
END IF;
```

```
UPDATE reservation01 SET TOTAL_COST = total;
UPDATE reservation01 SET END_TIME = end_hour;
```

```
RETURN total;
```

```
END;
```

```
/
```

```
DECLARE
```

```
cost NUMBER;
```

```
BEGIN
```

```
cost := CalcCost(3, 1);
```

```
DBMS_OUTPUT.PUT_LINE(cost);
```

```
END;
```

```
/
```

B. Develop a procedure named ListReservations that would list all the reservations for a specific bike, when the bike_ID is passed as the parameter. Make sure to list half day or full day in each reservation[4 marks]

```
Rent Details for the Bike B0001
C0001 8 14 10-OCT-24 Half Day 11.25
C0002 16 22 10-OCT-24 Half Day 11.25
```

```
CREATE OR REPLACE PROCEDURE ListReservations
(B_ID CHAR)
IS BEGIN
  CURSOR resC is
    SELECT R_ID, START_TIME, END_TIME, date_of_rent,
      CASE
        WHEN (END_TIME - START_TIME) >06 THEN 'full day'
        ELSE 'part day'
      END AS duration,
      CASE
        WHEN (END_TIME - START_TIME) >= 6 THEN cost_per_day
        ELSE (cost_per_day / 2)
      END AS cost
  END;
/
```

C. Create a trigger that will display a message when reservation are made. The criteria is :

1. Check if the client already had the booking on the same date. If yes – write message ‘Welcome back! Thanks for coming back today’
If No – ‘Welcome to our rental shop’ [3 marks]

```
CREATE OR REPLACE TRIGGER checkBooking AFTER INSERT ON reservation01
FOR EACH ROW
DECLARE count NUMBER;

BEGIN
  SELECT COUNT(*) INTO count FROM reservation01 WHERE

END;
/
```

Step 3. Now write the SQL commands to do the following:[1+1+1+1 marks]

- a) List **all** bikes and their total revenues from rentals. For bikes without rentals, the total charge should be 0. List the results from the highest revenue to the lowest revenue. Here is a sample listing:

Bike_ID	Total Revenue
B0001	600
B0002	149
B0003	0
B0004	0

```
SELECT Bike_ID, Total_Revenue FROM bikes b
LEFT JOIN reservation01 r ON b.B_ID = r.B_ID
GROUP BY b.B_ID
ORDER BY Total_Revenue DESC;
```

- b) **Add a new column** to table bike, which will store images of the bike(large jpg files).

```
ALTER TABLE bikes
ADD Image BLOB;
```

- c) List all the reservations for today with bike details

```
SELECT * FROM reservation01 WHERE (date_of_rent = SYSDATE);
```

- d) What command would you write to make sure all the work done today is saved permanently.

```
COMMIT;
```